1. Introduction and main results

Conclusions

2. Preliminaries

Definition 2.1 (Multifold sums of powers)**.** *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

Definition 2.2 (Central factorials)**.** *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n\left(n + \frac{k}{2} - 1\right)\left(n + \frac{k}{2} - 2\right)\cdots\left(n - \frac{k}{2} + 1\right) = n\left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1}\left(n + \frac{k}{2} - j\right)$ are falling factorials.

Proposition 2.3 (Central Newton's formula)**.** *For arbitrary integers x, and t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes central finite difference of $f(t)$, defined by

$$\delta^k f(t) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} f(t + \tfrac{k}{2} - j).$$

Proof. See [**?**, p. 462]. □

Lemma 2.4 (Binomial form of central factorials)**.** *For integers n and $k \geq 1$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n + \frac{k}{2} - 1}{k - 1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!}\left(n+\frac{k}{2}-1\right)_{k-1} = \frac{n}{k(k-1)!}\left(n+\frac{k}{2}-1\right)_{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

Lemma 2.5 (Halved central factorials). *For integers $n, k \geq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\binom{n+\frac{k}{2}}{k} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k}.$$

Proof. We have $\binom{a}{k} = \frac{a}{k}\binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n+\frac{k}{2}}{k} = \frac{n+\frac{k}{2}}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} + \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

Thus,

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \binom{n+\frac{k}{2}}{k} - \frac{1}{2}\binom{n+\frac{k}{2}-1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k}\binom{n+\frac{k}{2}-1}{k-1} = \frac{1}{2}\left[2\binom{n+\frac{k}{2}}{k} - \binom{n+\frac{k}{2}-1}{k-1}\right] = \frac{1}{2}\left[\binom{n+\frac{k}{2}}{k} + \binom{n+\frac{k}{2}-1}{k} + \binom{n+\frac{k}{2}-1}{k-1} - \binom{n+\frac{k}{2}-1}{k-1}\right].$$

Thus, the statement follows. □

Proposition 2.6 (Central Newton's formula for powers). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!}\delta^k t^m.$$

Proposition 2.7 (Halved central factorial powers). *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}}{k} + \binom{n-t+\frac{k}{2}-1}{k}\right]\delta^k t^m$$

Proposition 2.8 (Generalized hockey-stick identity). *For integers a, b and j,*

$$\sum_{k=a}^{b}\binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^{b}\binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right)$. By hockey stick identity $\sum_{k=0}^{n}\binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^{b}\binom{k}{j} = \left(\sum_{k=0}^{b}\binom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 2.9 (Centered hockey-stick identity)**.** *For integers $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Proof. By setting $j = 1-t+\frac{k}{2}+r$, and $b = n-t+\frac{k}{2}+r$ yields

$$\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r}\binom{j}{k}.$$

By Generalized hockey-stick identity (2.8),

$$\sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r}\binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

Thus, the claim follows. □

Proposition 2.10 (Ordinary sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\delta^k t^m \sum_{j=1}^{n}\left[\binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k}\right]$$

Proposition 2.11 (Ordinary sums of powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1}\right]\delta^k t^m$$

Proposition 2.12 (Central factorial numbers sums)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n+\frac{k}{2}+1}{k+1} - \binom{1+\frac{k}{2}}{k+1} + \binom{n+\frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1}\right] k! T(m,k),$$

where $T(n,k) = \frac{\delta^k 0^m}{k!}$ denote Central factorial numbers of the second kind (in Riordan's notation, see [**?**, p. 213]*, and* [**?**]*.), defined by central Newton's formula for powers, evaluated in $t = 0$*

$$n^m = \sum_{k=0}^{\infty} T(m,k) n^{[k]}.$$

Definition 2.13 (Generalized Central factorial numbers of the second kind)**.** *For integers $0 \leq k \leq m$, and an arbitrary integer t,*

$$T_t(m,k) = \frac{\delta^k t^m}{k!},$$

defined by central Newton's formula for powers, evaluated at arbitrary integer t

$$n^m = \sum_{k=0}^{\infty} T_t(m,k)(n-t)^{[k]}.$$

Proposition 2.14 (Generalized Central factorial sums)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1}\right] k! T_t(m,k)$$

Proposition 2.15 (Double sums of powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\begin{aligned}\Sigma^2 n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[&+ \binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2}\Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+1}\Sigma^1 n^0 \\ &+ \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2}\Sigma^0 n^0 - \binom{0-t+\frac{k}{2}}{k+1}\Sigma^1 n^0 \Bigg] \delta^k t^m\end{aligned}$$

Proposition 2.16 (Triple sums of powers decomposition)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^3 n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg[+ \binom{n-t+\frac{k}{2}+3}{k+3} - \binom{3-t+\frac{k}{2}}{k+3}\Sigma^0 n^0 - \binom{2-t+\frac{k}{2}}{k+2}\Sigma^1 n^0 - \binom{1-t+\frac{k}{2}}{k+1}\Sigma^2 n^0 + \binom{n-t+\frac{k}{2}+2}{k+3} - \binom{2-t+\frac{k}{2}}{k+3}\Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+2}\Sigma^1 n^0 - \binom{0-t+\frac{k}{2}}{k+1}\Sigma^2 n^0\Bigg]\delta^k t^m$$

Proposition 2.17 (Multifold sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0\Bigg\}\delta^k t^m$$

Lemma 2.18 (Multifold sum of zero powers)**.** *For integers $r \geq 0$ and $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

Proof. (1) Let $r = 0$, then $\Sigma^0 n^0 = n^0 = \binom{n-1}{0} = 1$, by definition (**??**).

(2) Let $r = 1$, then $\Sigma^1 n^0 = \sum_{k=1}^{n}\binom{k-1}{0} = \sum_{k=1}^{n} 1 = \binom{n}{1}$.

(3) Let $r = 2$, then $\Sigma^2 n^0 = \sum_{k=1}^{n}\binom{k}{1} = \sum_{k=1}^{n} k = \binom{n+1}{2}$.

(4) Let $r = 3$, then $\Sigma^3 n^0 = \sum_{k=1}^{n}\binom{k+1}{2} = \binom{n+2}{3}$.

(5) By induction over r and hockey stick identity $\sum_{k=r}^{n}\binom{k}{r} = \binom{n+1}{r+1}$, the claim follows $\Sigma^r n^0 = \binom{r+n-1}{r}$.

□

Proposition 2.19 (Multifold binomial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\Bigg\}\delta^k t^m$$

Proposition 2.20 (Multifold central factorial sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\Bigg\{\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\binom{s+n-1}{s}\Bigg\}k!T_t(m,k)$$

Definition 2.21.

$$R(n,t,k,r) = \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} - \sum_{s=0}^{r-1}\left[\binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s}\right]\Sigma^s n^0$$

Proposition 2.22 (Multifold sums of powers)**.** *For integers $n, m \geq 0$, and an arbitrary integer t,*

$$\Sigma^r n^m = \sum_{k=0}^{m}\frac{\delta^k t^m}{2}R(n,t,k,r)$$

Example 2.23 (Sums of squares).

$$t=0: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(1+n) + \frac{\delta^2 t^2}{2}\, \frac{n(1+3n+2n^2)}{6},$$

$$t=1: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-1) + \frac{\delta^2 t^2}{2}\, \frac{n(1-3n+2n^2)}{6},$$

$$t=2: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-3) + \frac{\delta^2 t^2}{2}\, \frac{n(13-9n+2n^2)}{6},$$

$$t=3: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-5) + \frac{\delta^2 t^2}{2}\, \frac{n(37-15n+2n^2)}{6},$$

$$t=4: \quad \Sigma^1 n^2 = \frac{t^2}{2}(2n) + \frac{\delta t^2}{2}\, n(n-7) + \frac{\delta^2 t^2}{2}\, \frac{n(73-21n+2n^2)}{6}.$$

Example 2.24 (Sums of cubes).

$$t=0: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(1+n) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(1+3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-1+n+4n^2+2n^3)}{24},$$

$$t=1: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-1) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(1-3n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(1+n-4n^2+2n^3)}{24},$$

$$t=2: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-3) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(13-9n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-21+25n-12n^2+2n^3)}{24},$$

$$t=3: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-5) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(37-15n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-115+73n-20n^2+2n^3)}{24},$$

$$t=4: \quad \Sigma^1 n^3 = \tfrac{t^3}{2}(2n) + \tfrac{\delta t^3}{2}\, n(n-7) + \tfrac{\delta^2 t^3}{2}\, \tfrac{n(73-21n+2n^2)}{6} + \tfrac{\delta^3 t^3}{2}\, \tfrac{n(-329+145n-28n^2+2n^3)}{24}.$$

Example 2.25 (Double sum of squares).

$$t=0: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(1+n)^2(2+n)}{12},$$

$$t=1: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-1+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n^2(-1+n^2)}{12},$$

$$t=2: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-4-3n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(10+5n-4n^2+n^3)}{12},$$

$$t=3: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-7-6n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(32+23n-8n^2+n^3)}{12},$$

$$t=4: \quad \Sigma^2 n^2 = \tfrac{t^2}{2}\, n(1+n) + \tfrac{\delta t^2}{2}\, \tfrac{n(-10-9n+n^2)}{3} + \tfrac{\delta^2 t^2}{2}\, \tfrac{n(66+53n-12n^2+n^3)}{12}.$$

Example 2.26 (Triple sum of squares)**.**

$$t=0: \quad \Sigma^3\, n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(6+11n+6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(18+45n+40n^2+15n^3+2n^4)}{120},$$

$$t=1: \quad \Sigma^3\, n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-2-n+2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(-2-5n+5n^3+2n^4)}{120},$$

$$t=2: \quad \Sigma^3\, n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-10-13n-2n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(58+65n-5n^3+2n^4)}{120},$$

$$t=3: \quad \Sigma^3\, n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-18-25n-6n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(198+255n+40n^2-15n^3+2n^4)}{120},$$

$$t=4: \quad \Sigma^3\, n^2 = \tfrac{t^2}{2}\,\tfrac{n(2+3n+n^2)}{3} + \tfrac{\delta t^2}{2}\,\tfrac{n(-26-37n-10n^2+n^3)}{12} + \tfrac{\delta^2 t^2}{2}\,\tfrac{n(418+565n+120n^2-25n^3+2n^4)}{120}.$$

Lemma 2.27 (Stirling form of Binomial)**.** *For integers n, k*

$$\binom{n}{k} = \sum_{j=0}^{k} \frac{(-1)^{k-j}}{k!} \begin{bmatrix} k \\ j \end{bmatrix} n^j$$

3. Mathematica

Mathematica Function	Validates / Prints
`MultifoldSumOfPowersRecurrence[r, n, m]`	Computes $\Sigma^r n^m$, (2.1)
`CentralFactorial[n, k]`	Computes $n^{[k]}$, (2.2)
`ValidateCentralFactorialsInTermsOfFalling[10]`	Validates Lem. (??)
`ValidateHalvedCentralBinomials[20]`	Validates Lem. (2.5)
`ValidateNewtonsFormulaForPowers[10]`	Validates Prop. (2.6)
`ValidateHalvedCentralFactorialPowers[5]`	Validates Prop. (2.7)
`ValidateCenteredHockeyStickIdentity[5]`	Validates Lem. (2.9)
`ValidateOrdinarySumsOfPowersDecomposition[10]`	Validates Prop. (2.11)
`ValidateDoubleSumsOfPowersDecomposition[5]`	Validates Prop. (2.15)
`ValidateTripleSumsOfPowersDecomposition[5]`	Validates Prop. (2.16)
`ValidateMultifoldSumsOfPowers[3]`	Validates Prop. (2.17)